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DUAL PATH SURFACE RECONSTRUCTION IN THE H/Ni(110) SYSTEM

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Reconstruction of a Ni(110) surface under the influence of adsorbed hydrogen atoms can proceed in two different ways: Below 180 K a 2x1 lattice-gas structure with Θ_H = 1.0 transforms cooperatively into a two-dimensional 1x2 structure by additional uptake of hydrogen up to Θ_H = 1.5. At higher temperatures, activated local transformation into a more stable one-dimensional structure starts already at low coverages.

The formation of ordered adsorbed layers in which the substrate lattice remains undistorted and the adsorbed particles are located at sites with identical local symmetry (lattice gas systems) has already been extensively investigated in the past[1]. Quite frequently, however, the atoms of the substrate surface are displaced from their original positions through formation of the chemisorption bond so that the surface reconstructs or an already existing reconstruction is removed [2-5]. In this article we report first evidence that such a process may proceed through two channels, leading to 2D and 1D ordered reconstructed phases, respectively. Since the rates of transformation exhibit different temperature dependence, these two situations could be well separated from each other. These observations were made with the H/Ni(110) system whose structural features had already been extensively studied in the past [6-10]. however, without recognition of the peculiar phase transformation properties. In addition, we offer an explanation for the occurrence of an extremely narrow peak in the thermal desorption spectra and present structure models for the two reconstructed surfaces which are consistent with all existing experimental data.

The experiments were performed between 120 K and 400 K by using Video-LEED, work function ($\Delta\Phi$), and thermal desorption (TDS) techniques after preparation of a clean Ni(110) surface in the usual manner[11].

Exposure of the clean Ni(110) surface to H_2 at 120 K leads to the formation of a continuous sequence of ordered lattice gas structures from adsorbed H-atoms which are completed by a 2x1 structure at Θ_H = 1.0 (Fig. 1a). Beyond this coverage a first-order phase transition into islands of a reconstructed 1x2 phase with local coverage Θ_H = 1.5 takes place, until the whole surface is converted into the latter phase[12]. The indicated coverage assignments were recently confirmed by nuclear microanalysis[13]. In the same work it was shown by Rutherford backscattering that the surface reconstructs by a pairing of adjacent parallel Ni rows in [001] direction in the topmost layer as first proposed by Tucker[14] and experimentally established by He diffraction[10] as well as by ion channeling[15]. So far the positions of the adsorbed H atoms were unclear, but we present a plausible structure model in fig. 1b. This model follows directly from the well established 'zig-zag' array of the H atoms in the 2x1 phase formed on the unreconstructed Ni(110) surface (fig. 1a)[10]. If two neighboring Ni rows are shifted together, every second adsorption site with threefold local symmetry is converted into a site with fourfold local symmetry (although it has to be kept in mind that the entire reconstructed system still has twofold symmetry). The remaining 'threefold' sites exhibit now a large enough mutual separation that they can accommodate additional H atoms leading to Θ_H = 1.5. This transformation proceeds by cooperative displacement of fairly long rows of Ni atoms, but is not associated with an appreciable activation energy barrier. This conclusion is supported by the observation that the sticking coefficient for dissociative adsorption remains fairly high even for $\Theta_H > 1.0$. A cooperative mechanism appears plausible, if the ordered 2x1 structure is a prerequisite for the transformation into the well-ordered 1x2 structure. The formation of the latter phase has never been observed without prior passage through the 2x1 phase and therefore does not occur at temperatures above its disordering at about 180 K. The proposed structure model is not only a consequence of the structure of the 2x1 phase and agrees with the experimentally derived coverage, but could also explain recent vibrational spectroscopic data: An energy loss of 74 meV observed with the 1x2 phase[16] is also present with the H/Ni(100) system[17] where the H atoms are most likely located in fourfold sites[17,18].

The situation is quite different if the clean surface is exposed to H_2 at higher temperatures, e.g., 300 K: Now at already fairly low coverages streaks become visible in the LEED pattern as sketched in Fig. 3b, indicating the formation of a one-dimensionally ordered surface phase which is again characterized by lateral displacement of Ni surface atoms[13]. These streaks are very sharp and initially very homogeneous in $[1\bar{1}0]$ direction. The

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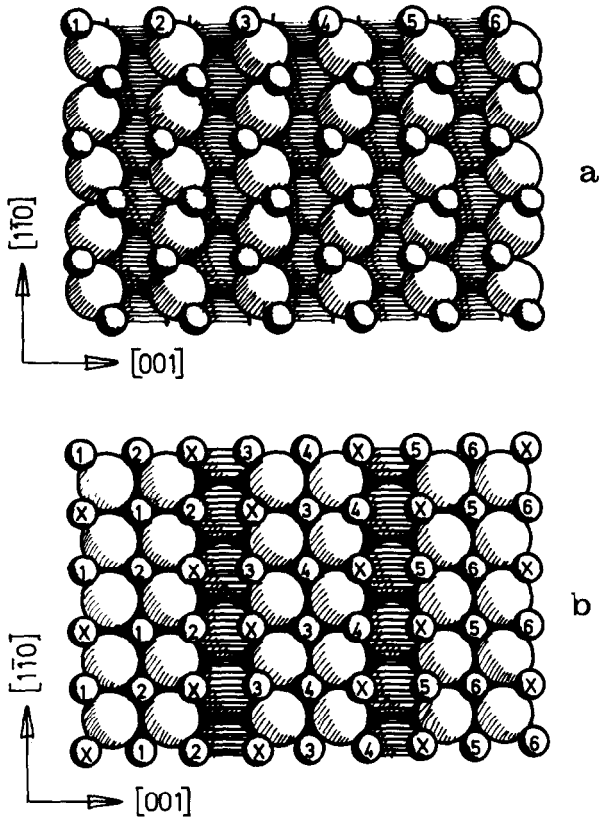


Figure 1 a) Structure model for the 2x1 phase formed with Θ_H = 1.0 at T < 180 K

b) Model for the reconstructed 1x2 phase with Θ_H = 1.5. Inserted numbers denote the H atoms originating from the preceeding 2x1 structure, while H atoms labelled with "X" are additionally accommodated

reconstruction into this phase only occurs above ~ 180 K.

Contrary to the 2x1 → 1x2 transition, formation of the streak phase does not start from an ordered H overlayer. If we assume in this case local instead of cooperative pairing of neighboring doublets of Ni atoms, we arrive at a structural model for the streak phase which is shown in fig. 3a. Since the actual location of the H atoms is not known (although threefold and fourfold sites are most likely occupied) we have only drawn the Ni atoms of the first two layers. The periodicity along the [1$\bar{1}$0] direction is still preserved while it is heavily distorted along the [001] direction. This 1D-structure would clearly give rise to the observed 'streak' LEED pattern of fig. 3b. Our model would also be consistent with the observation that the average displacement of Ni surface atoms is smaller with the streak phase than with the 1x2 phase[13]. If a surface covered with Θ_H<1.0 at 120 K is continuously heated up, the ordered H lattice gas structures disorder around 180 K and the streaks described above begin to form. The intensity of the streaks rises up to ~ 320 K (where complete desorption occurs) indicating that this transformation is thermally activated (cf. fig. 2b). The thermal desorption data exhibit two states, β_1 and β_2 (fig. 2a), which are simultaneously populated and whose intensity ratio depends on the heating rate dT/dt[11]. Rates between 1 K/s and 25 K/s have been used in our experiments. At high dT/dt relatively more H_2 molecules come off in the β_1-state. We attribute this latter state to desorption from the non-reconstructed 1x1 surface with desorption energy $E_d(\beta_1)$, whilst the β_2-state arises from desorption from the reconstructed streak phase. The energy difference $E_d(\beta_2) - E_d(\beta_1)$ is obviously the driving force for the 1x1 → → streak transformation whose activation energy $E^*_{tr} < E_d(\beta_1)$ - hence the competition between β_1-desorption and reconstruction during heating up, where the former process gains more weight at higher heating rates and vice versa.

If the 1x2 phase is continuously heated up, it decomposes suddenly in a narrow temperature range around 220 K as rationalized by the experimental data depicted in fig. 2: About half a monolayer of H atoms desorbs in a narrow α-state (fig. 2a) accompanied by a steplike decrease of the work function as well as of the intensity of the half-order spots in the 1x2 LEED pattern (fig. 2b). This desorption state had already been reported previously where, however, no satisfactory explanation could be offered[19]. A similar effect was observed in connection with a catalytic decomposition reaction and was denoted by the term 'surface explosion'[20]. Simultaneously the intensity of the integral-order (1,0) LEED spots rises (fig. 2b). The kinetics of the desorption process in the α-state is obviously not governed

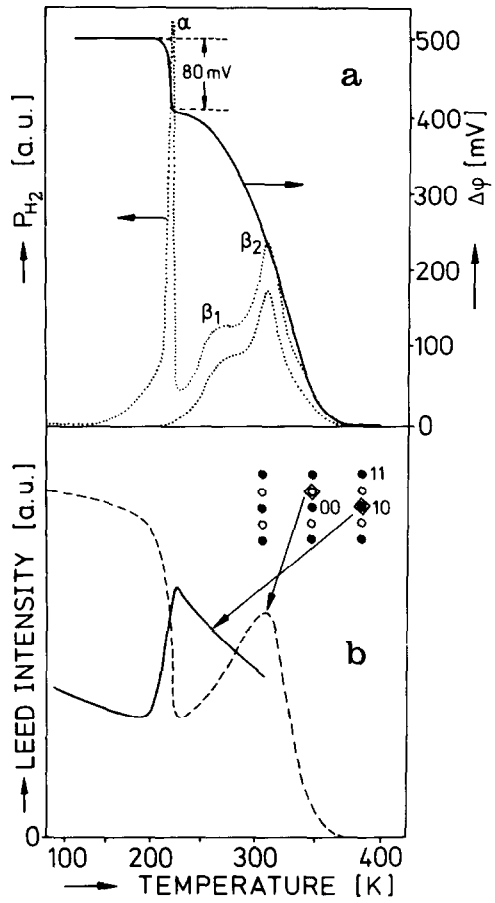


Figure 2 Annealing of a H covered Ni(110) surface, exposed to hydrogen at 120 K:
a) TDS ($\Theta_H < 1.0$, $\Theta_H = 1.5$) and $\Delta\varphi$ ($\Theta_H = 1.5$)
b) LEED intensity ($\Theta_H = 1.5$):
···· half-order spot; ——— (1,0) spot

by recombination of adsorbed H atoms, but rather by the decomposition (=first order transition) of the 1x2 domains, associated with relaxation into the non-reconstructed surface structure. As soon as the cooperative process is completed, the hydrogen desorption stops, which offers a natural explanation of the narrow temperature width of the α-state. While the fraction of H atoms exceeding $\Theta_H = 1$ desorbs in the α-state, the remainder again shows up as β_1 and β_2-states in the TD spectrum.

In the parallel LEED experiment the electron energy was chosen in such a way that the intensity of the (1,0) spot is high with the non-reconstructed surface, but low with the two reconstructed phases [11]. The resulting intensity increase at 220 K thus indicates that, during desorption of the α-state, the surface may relax back into the non-reconstructed 1x1 structure. This argument is supported by the observation that rapid cooling of the sample after desorption of the α-state leads to a partial restoration of the 2x1 phase with $\Theta_H = 1.0$, which exists on the non-reconstructed Ni surface only. The H covered surface thus has two pathways for reconstruction: Into the 1x2 and into the 'streak' structure. It has been demonstrated that there is no transition from the 1x2 into the 'streak' structure, rather after release of the H atoms in excess of $\Theta_H = 1$ the surface converts into the 1x1 phase which then gradually transforms into the 'streak' structure.

The first path, reconstruction into the 1x2 structure at $\Theta_H > 1$ and $T < 180$ K, obviously starts from an ordered H overlayer and exhibits negligible activation energy. In contrast the local pairing process in path II - reconstruction into the 'streak' phase - evidently requires an appreciable activation energy. Unlike the 1x2 phase, desorption from the streak phase will in turn not be a cooperative phenomenon which accounts for the 'normal' shape of the β_2-desorption state. With respect to the total energy, the 'streak phase' is the most stable one ($\triangleq \beta_2$). In turn the 1x2 ($\triangleq\alpha$) and the 2D-lattice gas structures ($\triangleq\beta_1$) are metastable and can only be observed because of the suppression of the 1x1 → 'streak' transformation at low temperatures. The energetic difference between the two reconstructed phases of at least 25 kJoule

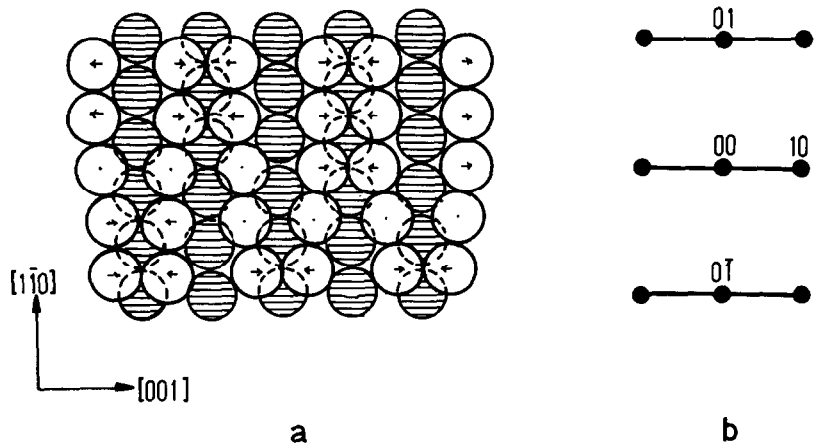


Figure 3 a) Proposed structure model of the 'streak' phase (Ni atoms shown only). The direction of distortion of the topmost Ni atoms, with respect to their bulk position, is indicated by arrows.

b) LEED pattern of the 'streak' phase, after prolonged exposure to hydrogen at room temperature.

/mol ($\alpha \leftrightarrow \beta_2$) seems too large to solely result from different H - H interactions; we attribute it to an energetically more favorable arrangement of the Ni atoms in the 'streak' structure as compared to the 1x2 phase.

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